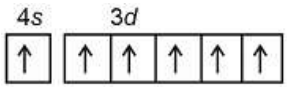
ANSWER KEY

AIATS for Regular Medical-2026(XII Passed)_Practice Test-1

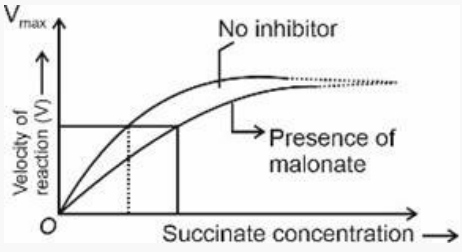
Q.No	Correct Option	Answer
1	D	Parabolic
2	B	8 m/s
3	C	A(S); B(P); C(Q); D(R)
4	C	(5.30 ± 0.05)
5	B	0.04%
6	D	$20 - 10\sqrt{3}\,\hat{i} + 10\hat{j}$
7	C	40.42 m/s
8	A	$\frac{x}{4}$
9	C	400000
10	B	$\frac{1}{9}$
11	B	$F^{3/2}v^{-1}d^{-1/2}$
12	C	$-4\ m/s^2$
13	A	3
14	B	$[LT^2]$
15	C	10%
16	C	5 m/s
17	B	5.77 s
18	B	Only statement B is correct
19	A	Both (A) and (R) are true and (R) is the correct explanation of (A)

20	**D**	3.75 hours
21	**C**	22.5 s
22	**A**	x
23	**C**	$20\sqrt{2}\ \mathrm{m/s}$
24	**B**	$\frac{5\sqrt{3}}{2}$, $\frac{5}{2}$
25	**D**	A(R); B(P); C(Q); D(S)
26	**C**	$M^{-1}L^{6}T^{2}$
27	**C**	(A) is true but (R) is false
28	**B**	$\sqrt{2}$
29	**B**	5 m/s
30	**C**	Rydberg constant and universal gas constant are dimensionally same
31	**D**	Both (1) and (3)
32	**B**	Both (A) and (R) are true but (R) is not the correct explanation of the (A)
33	**D**	120 m
34	**D**	1 sec
35	**B**	$S_2 = 4S_1$
36	**C**	$\frac{9}{2}$
37	**D**	Both statement I and statement II are incorrect
38	**A**	Velocity
39	**D**	Both (2) and (3)
40	**B**	$\frac{16}{3}\ \mathrm{km}\ /\ \mathrm{hr}$
41	**C**	$\frac{2}{10\ \mathrm{N}}$
42	**B**	$\frac{19}{5}\mathrm{m}$
43	**C**	$250\sqrt{2}\ \mathrm{km}$
44	**C**	$\frac{1}{\sqrt{3}}\mathrm{s}$
45	**B**	3 m/s
46	**D**	b and d only

47	**A**	20.5 u
48	**A**	1 : 1
49	**B**	5 g of H_2 gas
50	**B**	(iv) > (iii) > (ii) > (i)
51	**C**	Increased by 2.24 L
52	**C**	0.2 g
53	**C**	Mg and 1.04 g
54	**D**	$6N_A$
55	**B**	0.88 g
56	**A**	5.9×10^5 u
57	**D**	0.1 M
58	**C**	a–(iv), b–(iii), c–(ii), d–(i)
59	**B**	Molarity of solution will be 0.06 M
60	**B**	0.98
61	**C**	0.266 M
62	**C**	0.4 M & 0.6 M
63	**A**	X_3O_4
64	**A**	2 mol
65	**D**	(4)
66	**B**	2.04×10^{20}
67	**D**	a < c < b < d
68	**D**	a – (iv), b – (ii), c – (iii), d – (i)
69	**C**	$r_4 = \frac{8}{9}r_3$
70	**B**	–13.6 eV
71	**C**	2.02×10^{18}
72	**D**	(a) only
73	**B**	7.96%

74	**D**	a–(iv), b–(iii), c–(ii), d–(i)
75	**D**	Spin quantum number
76	**D**	Statement I is correct but statement II is incorrect
77	**A**	Both (A) and (R) true and (R) is the correct explanation of (A)
78	**A**	656 nm
79	**A**	The oscillating electric and magnetic fields produced by oscillating charged particles are parallel to each other and both are perpendicular to the direction of propagation of wave
80	**A**	13 : 19
81	**B**	
82	**B**	Radial node = 1
83	**C**	Total number of unpaired electrons in Cr atom and Fe atom are same
84	**B**	2.4×10^{-12} m
85	**D**	n = 2, ℓ = 2, m = –1, s = $-\frac{1}{2}$
86	**C**	a–(iv), b–(iii), c–(ii), d–(i)
87	**C**	n = 5 to n = 1
88	**C**	4
89	**A**	Lyman
90	**A**	(i) & (ii)
91	**B**	Only a and d are correct
92	**C**	(3)
93	**D**	The inner compartment is filled with dense homogenous substance called matrix.
94	**C**	Lipids and protein
95	**A**	Appears to have a single arm.
96	**C**	Both (c) and (d)
97	**A**	18S rRNA
98	**D**	Only (a) and (d)

99	**A**	Terminalisation of chiasmata
100	**B**	A-(F), B-(F), C-(F), D-(T), E-(T)
101	**C**	Chromosomes
102	**C**	Dissimilar to each other as well as to the parent cell.
103	**B**	a–(iii), b–(ii), c–(iv), d–(i)
104	**C**	Lion, tiger and leopard
105	**D**	Carolus Linnaeus
106	**D**	Acellular organisation
107	**D**	Species
108	**C**	A-(F), B-(F), C-(T), D-(F)
109	**D**	Both statements A and B are incorrect
110	**A**	Ribosomes – They are associated with plasma membrane in prokaryotes
111	**A**	a only
112	**D**	Congression of chromosomes can be observed
113	**B**	c → b → d → a
114	**C**	d only
115	**B**	Both statement A and B are false.
116	**D**	a–(iii), b–(iv), c–(ii), d–(i)
117	**B**	a, b and e are correct
118	**B**	27 and 20
119	**C**	Glycocalyx
120	**D**	Presence of tonoplast
121	**A**	Ernst Mayr
122	**A**	a-(T), b-(F), c-(T)
123	**D**	*Cuscuta*
124	**D**	Mammalia and Insecta
125	**D**	(A) is true but (R) is false

126	**B**	Histone protein synthesis
127	**B**	Both (A) and (R) are true but (R) is not the correct explanation of (A)
128	**D**	They are bound by a single membrane.
129	**C**	(3)
130	**D**	Wheat, monocot, and plants represent taxa at the same level in the taxonomical hierarchy
131	**B**	Anaphase II
132	**C**	Systematics, Latin and systema
133	**D**	Both statements A and B are incorrect
134	**A**	20, 60
135	**C**	Anaphase – Shortening of spindle fibres that are attached to the kinetochore
136	**B**	Four
137	**B**	Perform cementing to keep the neighbouring cells together
138	**B**	Link and support other tissues/organs of the body
139	**A**	Cl_3CCOOH
140	**B**	Collagen
141	**A**	Intestine
142	**A**	Each skeletal muscle is made of many long, cylindrical fibres arranged in parallel arrays.
143	**B**	Four
144	**A**	
145	**C**	Only (a) and (b)
146	**D**	Four
147	**A**	Orientation of fibres
148	**C**	Every enzyme has a primary structure, *i.e.*, linear amino acid sequence of the protein.

149	**A**	Glycine
150	**C**	Only statement A is correct
151	**D**	Uninucleated and unbranched
152	**C**	Lyases
153	**A**	(a) and (b)
154	**C**	When cell membranes break into pieces, water-insoluble vesicles are formed which gets separated along with the acid-insoluble part
155	**D**	The amino acid located on the C-terminal of the chain possesses aromatic ring in its structure.
156	**D**	Only statement B is correct
157	**A**	Both (A) and (R) are true but (R) is not the correct explanation of (A)
158	**D**	Provides positional information of amino acids
159	**D**	Cylindrical and multinucleated
160	**C**	Two
161	**D**	Hexose sugar
162	**B**	(2)
163	**D**	Cube-like cells with microvilli to increase surface area for absorption
164	**D**	Myofibrils
165	**D**	Dense irregular connective tissue – Present in the skin
166	**C**	(a), (b) and (c) only
167	**B**	Goblet cells
168	**D**	(a), (b) and (c)
169	**B**	Lyases
170	**C**	Carbonic anhydrase markedly accelerates the reaction of carbonic acid into carbon dioxide and water by 1 million times relative to the uncatalyzed reaction.
171	**C**	Both (A) and (R) are false
172	**C**	Inner lining of pancreatic duct
173	**C**	Two

174	**C**	Less than one-half of the volume of neural tissue in our body
175	**D**	Morphine
176	**C**	Cytosine
177	**C**	Provides protection against chemical stresses.
178	**B**	Zinc
179	**C**	a(ii), b(iii), c(i), d(v), e(iv)
180	**C**	Former is composed of collagen fibres, calcium salts and osteocytes